



IMAGING: Ashwin Boricha

BEHIND THE MOVIES

Discover the magic that makes the latest movies larger than life

“You can’t rush art”, the old man repairing Woody tells an impatient toy collector in *Toy Story II*. Likewise, the spellbinding effects that we see on the big screen today—be it *The Matrix*, *Jurassic Park* or *Shrek*—take hundreds of manhours to craft. Many events such as the sinking of the Titanic or the simulation of tornadoes and explosions, now take place in the studios.

When Hollywood was young, special effects were spun by stop-motion animation and optical printing. An optical printer is a camera and projector operating together to take a photograph. Optical printers and various rear and front projection techniques were used to merge live action with backgrounds. For instance, in 1933, special effects pioneer Willis O’Brien brought the King Kong model to life by photographing it frame by frame, while optical printers and various compositing techniques were used to merge the gorilla with backgrounds and actors.

Computer animators now create more detailed and complex characters with 3D modelling software. The movement too looks more natural as computers can accurately control

frame blurring and blending. Besides, high-resolution scanners have replaced movie optical printers. Once digitised, shots can be composited, enhanced or worked on by digital animators and artists. After the special effects are completed, the footage is scanned back to 35 mm negative and is ready for printing and projection.

Take the case of *The Lord of the Rings*—Peter Jackson and his team from WETA

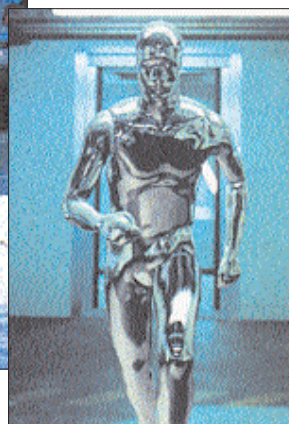
Digital, a New Zealand based FX company, not only created a Middle-earth but also designed an entirely digital universe for *The Lord of the Rings* trilogy with digital creatures such as Gollum, Treebeard, Balrog and the eye of Sauron. The transformation of Gollum—from a hobbit-like creature named Smeagol to something scarier through his own encounter with the Ring—was done by a combination of computer animation and sophisticated motion-capture technology using fluid dynamics.

To gather motion capture data, an actor is fitted with a suit that has reflective markers or lights at every joint. The actor moves on a special stage and 3D cameras watch the actor from a number of different angles. Computer software then tracks all of the markers and, with the help of a technician, bind them together into a stick figure that accurately duplicates the motion of the actor. The concept of fluid dynamics governs everything from airflow over a wing to any type of moving water.

Peter Jackson wanted a natural look, so Gollum’s realistic joint movement is based on actual organic muscle and bone, seen rippling under his translucent, but flesh-like skin. The computer artists even studied anatomy books to



For the film, *The Terminator*, black-grid painted Robert Patrick was filmed and the reference footage was scanned



The result: A T-1000 animation